Existing PIM Systems Research

Project scope: Standardization of product quantities and units

Group members: Yakup, Jeveria, Göktan, Sajad, Miran & Patritsia

Version: 1.0

Status: Research completed – Available Product Analysis

1. Research Goal

The goal of this research is to understand how existing Product Information Management (PIM) systems handle product data from external sources such as suppliers.

The research focuses on how product data is imported, structured, transformed, and stored inside a PIM system. It also looks at how quantities and units are represented.

The findings will be used to get an idea of where BestDeal's standardization solution could be integrated into the existing product data flow.

2. Research Method

For this research, the Available Product Analysis method was used.

This method focuses on examining existing products or solutions that already solve the same problem, or part of the problem. The purpose is to learn from existing solutions instead of designing a solution without first investigating what is already available.

Two existing PIM systems were selected for this analysis:

- **Akeneo**
- **Pimcore**

Akeneo provides official documentation describing how external product data can be imported into the PIM and how products are structured using elements such as attributes, families and categories. (Import Your Data, z.d.
- Concepts & Resources | Akeneo APIs, z.d.)

Pimcore provides official documentation describing how external product data can be imported, mapped to product fields and transformed before it is

stored. (Data Importer | PIMCore Development Documentation, z.d. - Mapping Configuration | PIMCore Development Documentation, z.d.)

The analysis focuses specifically on features that are relevant to the BestDeal project, such as:

- importing product data from external sources;
- structuring product attributes;
- handling quantities and units;
- transforming incoming product data;
- and identifying possible integration points for a standardization solution.

3. How Does a PIM Work?

A Product Information Management system is used as a central place for managing product information. It helps organizations keep product data consistent and available for different systems and sales channels. (Pimcore, 2025)

Product information can come from different external sources, such as suppliers or other business systems. After the data enters the PIM, it can be organized into a structured product model. In Akeneo, for example, products are structured using elements such as attributes, families and categories. (Concepts & resources | Akeneo APIs, z.d.)

The information stored in a PIM can then be managed, improved and distributed to other systems or channels, such as a webshop. Pimcore, for example, describes its PIM as a central system that manages product information and distributes it across different channels and systems. (Pimcore, 2025)

A simplified product data flow can therefore be represented as:

External data sources
↓
PIM
↓
Webshop or other channels

The exact way in which product data is imported, structured and processed depends on the PIM system. Therefore, Akeneo and Pimcore are analysed separately in the following sections.

4. Akeneo

Akeneo is an existing Product Information Management system. It allows product information from external sources to be imported and managed in a structured catalog.

4.1 Importing Product Data

Akeneo can import product information from external sources such as vendors and ERP systems. During an import, Akeneo extracts the data from the provided file and converts it into a format that can be saved in the PIM. CSV and XLSX files are supported. (Import Your Data, z.d.)

A simplified flow is:

External source
↓
Product file
↓
Import
↓
Akeneo PIM

4.2 Structuring Product Data

Inside Akeneo, product information is structured using attributes, families and categories.

An **attribute** represents a characteristic of a product, such as a name, identifier, price, number or measurement.

A **family** acts as a template that defines which attributes belong to a certain type of product.

A **category** is used to classify products into groups. (Concepts & resources | Akeneo APIs, z.d.)

4.3 Handling Quantities and Units

Akeneo has a specific Measurement attribute for information that contains both a numerical value and a measurement unit.

For example:

Value: 500
Unit: gram

Measurement units that belong to the same type are grouped into a Measurement Family. Examples of measurement families are weight and length. (What about measurements?, z.d.)

For each measurement family, Akeneo uses a Standard Unit. Other units in that family can be converted to the standard unit using defined conversion rules.

The standard unit is not required to be the same for every organization. Akeneo allows organizations to create their own measurement families, add measurement units and configure the units that are needed for their catalog. (What about measurements?, z.d.)

For example, an organization could decide to use millilitres as its standard representation for volume:

0.33 L
↓
330 ml

5. Pimcore

Pimcore is a platform that can be used as a Product Information Management system. Product information is stored as structured Data Objects, which can represent products and other business entities. (Data Objects | PIMCore Development Documentation, z.d.)

5.1 Importing Product Data

Pimcore provides a Data Importer for importing data from external sources.

It supports different data sources and formats, including CSV, JSON, XML and XLSX. (Data Importer | PIMCore Development Documentation, z.d.)

A simplified flow is:

External source
↓
Data Importer
↓
Pimcore Data Object

5.2 Mapping Product Data

During an import, Pimcore uses Mapping to determine where incoming data should be stored.

For example:

supplier_product_code → SKU
supplier_product_name → Name
supplier_weight → Weight

Each mapping connects data from the source to a field in the Pimcore Data Object. (Mapping Configuration | PIMCore Development Documentation, z.d.)

5.3 Transforming Incoming Data

Pimcore can transform incoming data before it is stored.

Its Transformation Pipeline takes source data and converts it into the format required by the target field. This can include converting data types or creating structured values such as a quantity combined with a unit. (Transformation Pipeline | PIMCore Development Documentation, z.d.)

The process can therefore be simplified as:

Source data
↓
Mapping
↓
Transformation
↓
Pimcore Data Object

6. Comparison and Findings

The analysis of Akeneo and Pimcore shows that both systems can structure product measurements, but they use different approaches. (Manage your measurements, z.d.; Number Datatypes | PIMCore Development Documentation, z.d.)

Akeneo	**Pimcore**
Uses **Measurement Attributes** to store a value together with a unit.	Uses **Quantity Value** to store a numerical value together with a unit.
Groups related units inside a **Measurement Family**.	Units can be configured centrally and connected through **base units**.
Uses a **Standard Unit** to convert other units in the same measurement family.	Uses **conversion factors** to convert between related units.
Allows organizations to create their own measurement families and units.	Allows organizations to configure the units and conversion relationships needed for their data model.

Both systems also provide ways to process incoming product data before it is stored.

Akeneo provides **Tailored Import**, which allows incoming supplier fields to be mapped to PIM attributes and transformed during the import process.

(Tailored import - import any file structure to Akeneo PIM, z.d.)

Pimcore provides **Mapping** and a **Transformation Pipeline**, which can map incoming fields to product fields and transform the data before it is stored.

(Mapping Configuration | PIMCore Development Documentation, z.d.; Transformation Pipeline | PIMCore Development Documentation, z.d.)

6.1 Findings for BestDeal

The research shows that there is no single required standard unit that every PIM must use. The way quantities and units are represented can be configured depending on the organization and the PIM system.

For BestDeal, suitable standard units could therefore be defined for different types of measurements. For example:

Volume → ml
Weight → g
Length → mm

However, these units should not be selected only by the project team. The most suitable standard units and quantity structure should be discussed and validated with BestDeal, based on their existing PIM, supplier data and business requirements.

The research also suggests that the standardization solution could be placed during the processing of incoming supplier data, before the standardized values are stored in the PIM.

A possible flow could therefore be:

Supplier feeds
↓
Import
↓
Mapping / Standardization
↓
Structured product data
↓
BestDeal PIM
↓
Webshop / fronted

The exact integration point still needs to be validated with BestDeal because the internal architecture and capabilities of their existing PIM have not yet been fully investigated.

7. Conclusion

The research shows that existing PIM systems such as Akeneo and Pimcore provide structured ways to import, organize and process product information.

Both systems support storing measurements as a combination of a value and a unit, but they use different structures and terminology.

Based on these findings, BestDeal could use a similar approach by standardizing supplier data during the import and processing stage before the standardized values are stored in the PIM.

The exact standard units, quantity structure and integration point should be discussed and validated with BestDeal because they depend on the company's existing PIM, supplier data and business requirements.

8. Sources

Akeneo

Akeneo. *Import your data*.
Import your data. (z.d.). Akeneo. https://help.akeneo.com/v7-import-your-data

Akeneo. *Catalog Structure*.
Concepts & resources | Akeneo APIs. (z.d.).
https://api.akeneo.com/concepts/catalog-structure.html

Akeneo. *What about measurements?*
What about measurements? (z.d.). Akeneo.
https://help.akeneo.com/en_US/serenity-discover-akeneo-concepts/serenity-what-about-measurements

Akeneo. *Manage your measurements*.
Manage your measurements. (z.d.). Akeneo. https://help.akeneo.com/en_US/v7-build-your-catalog/v7-manage-your-measurements

Akeneo. *Tailored Import – Map any file structure*.
Tailored import - import any file structure to Akeneo PIM. (z.d.). Akeneo.
https://help.akeneo.com/en_US/import-export-data/tailored-import-map-any-file-structur

Pimcore

Pimcore. *Product Information Management*.
Pimcore. (2025, 24 september). Product Information Management (PIM) | PIMCore. Pimcore. https://pimcore.com/en/products/product-information-management

Pimcore. *Data Objects*.
Data Objects | PIMCore Development Documentation. (z.d.). https://docs.pimcore.com/platform/Pimcore/Objects/

Pimcore. *Data Importer*.
Data Importer | PIMCore Development Documentation. (z.d.). https://docs.pimcore.com/platform/next/Data_Importer/

Pimcore. *Mapping Configuration*.
Mapping Configuration | PIMCore Development Documentation. (z.d.). https://docs.pimcore.com/platform/next/Data_Importer/Configuration/Mapping_Configuration/

Pimcore. *Transformation Pipeline*.
Transformation Pipeline | PIMCore Development Documentation. (z.d.). https://docs.pimcore.com/platform/Data_Importer/Configuration/Mapping_Configuration/Transformation_Pipeline/

Pimcore. *Number Datatypes – Quantity Value and Unit Conversion*.
Number Datatypes | PIMCore Development Documentation. (z.d.). https://docs.pimcore.com/platform/2026.1/Pimcore/Objects/Object_Classes/Data_Types/Number_Types/